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École polytechnique de Louvain

ÉCOLE POLYTECHNIQUE DE LOUVAIN

LEPL2022 - Health Innovation Classes

Introduction – Design Thinking – 3 pillars

Empathy - Hackathon

February 3rd, 2026

Pierre Geens, MME, ieMBA

Health Innovation Classes

Health Innovation Classes

- 5 credits
- Master students
- Multidisciplinary

Objectives

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Health

Beyond treatment…

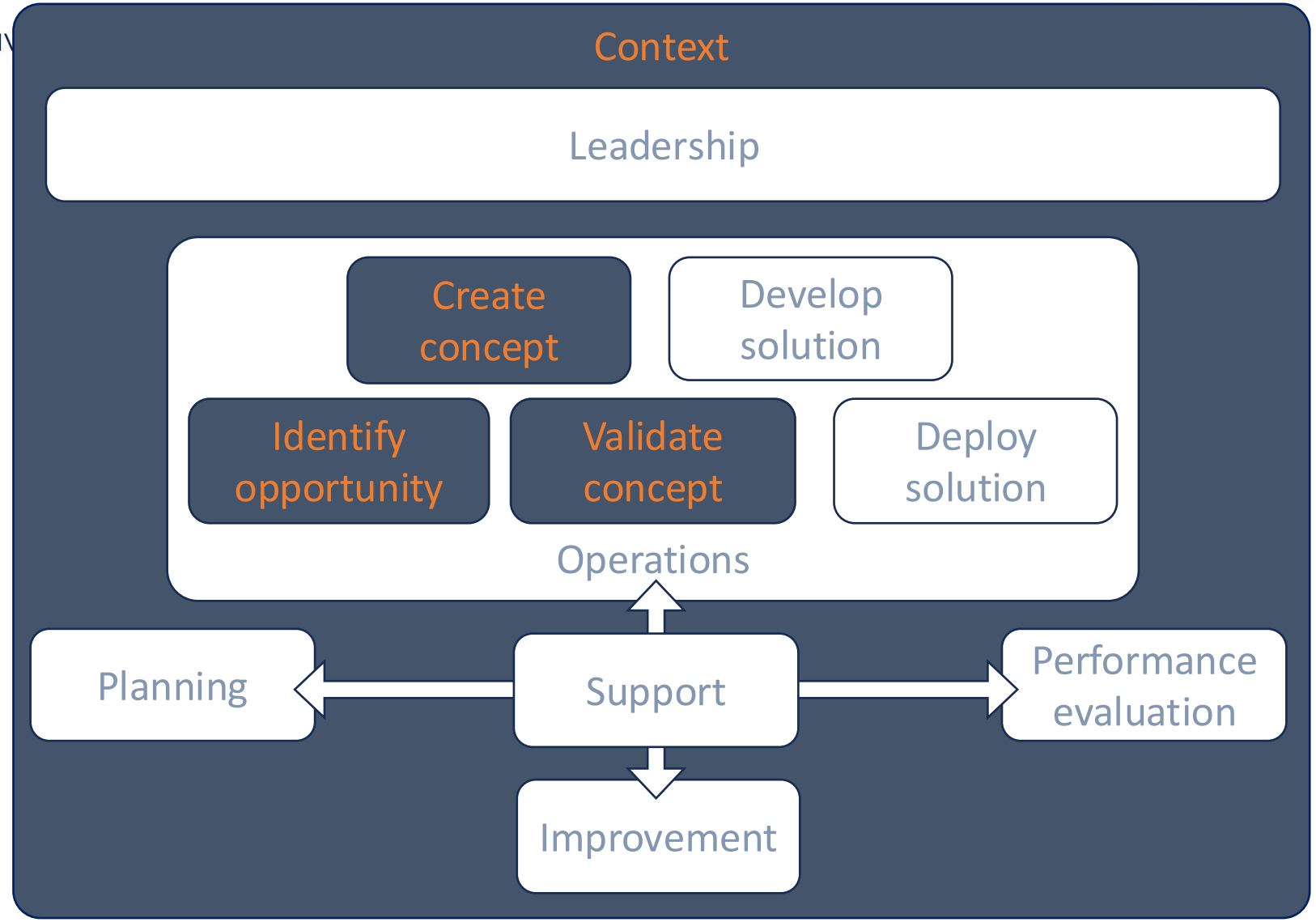
- Prevention
- Monitoring
- Personalized medicine
- Patient empowerment
- Efficiency

Innovation

Beyond idea generation…

- Idea evaluation
- Development
- Implementation
- Strategy
- Management

Specific focus on generating **valid concepts** that address pressing challenges within the **current context**



Learning objectives

Positioning project in regard to global healthcare context and challenges	Developing concept using ideation techniques	Assessing, iterating and validating concept against healthcare specific factors	Convincing about concept viability	Recognizing, leveraging and integrating value of peers from other disciplines

Modalities

©UCLouvain – Credit Jacky Delorme

70%

Practice

1,5 days hackaton
Group project
Field visits (hospital, industry,...)
Final pitch

Empathize with stakeholders
Ideation
Financing healthcare
Regulatory and legal requirements
Business planning

Seminars

30%

3 fields

Multi Disciplinary

Engineering
Health Sciences
Management

Who are you?

WHO ARE YOU?

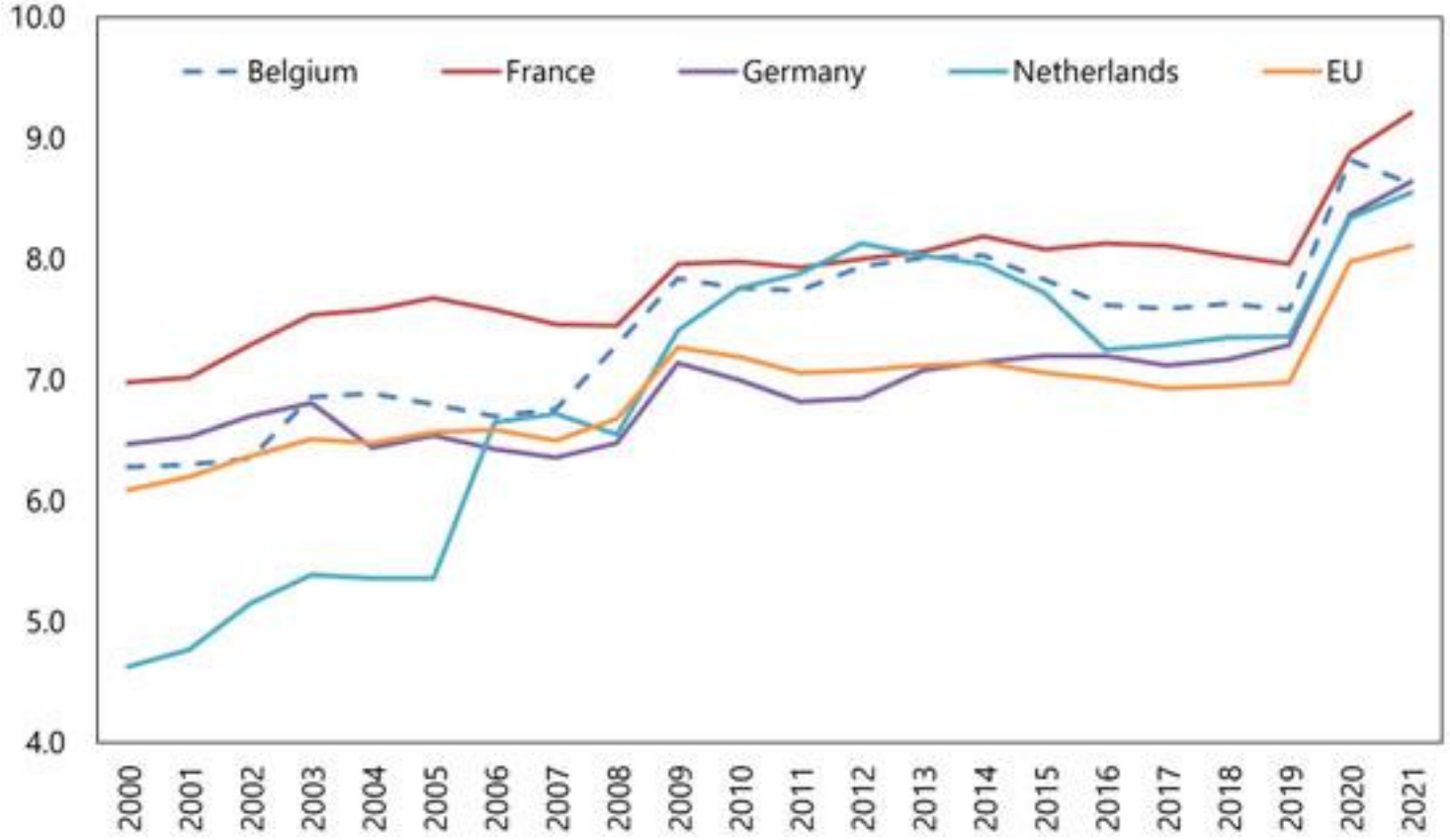
5mins - In groups of 3

- Name and surname
- Identify something you all have in common, related to health (e.g. we all had a bone broken)
- Identify, for each individual, something unique vs the rest of the group

20mins - Each group presents the outcome

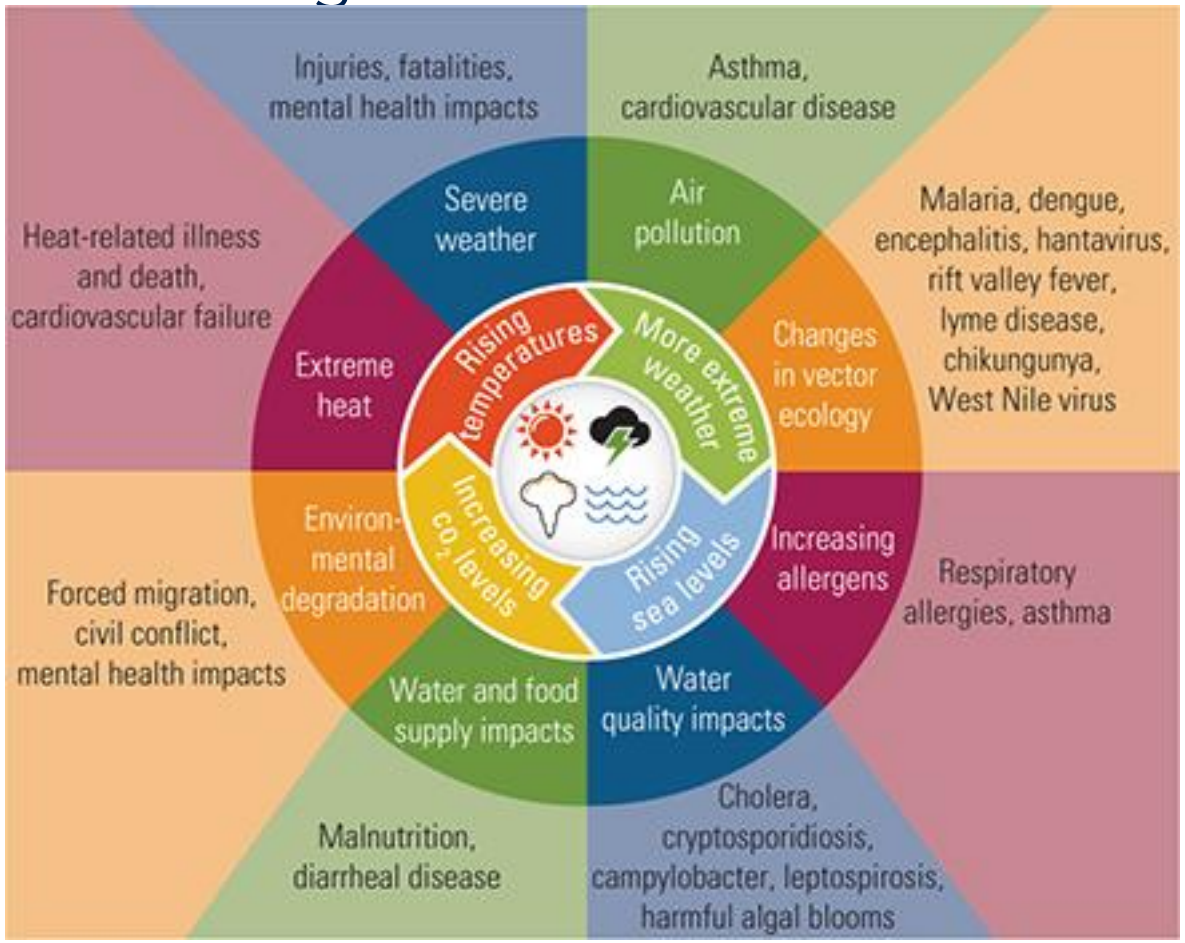
Innovation

The cost of ageing



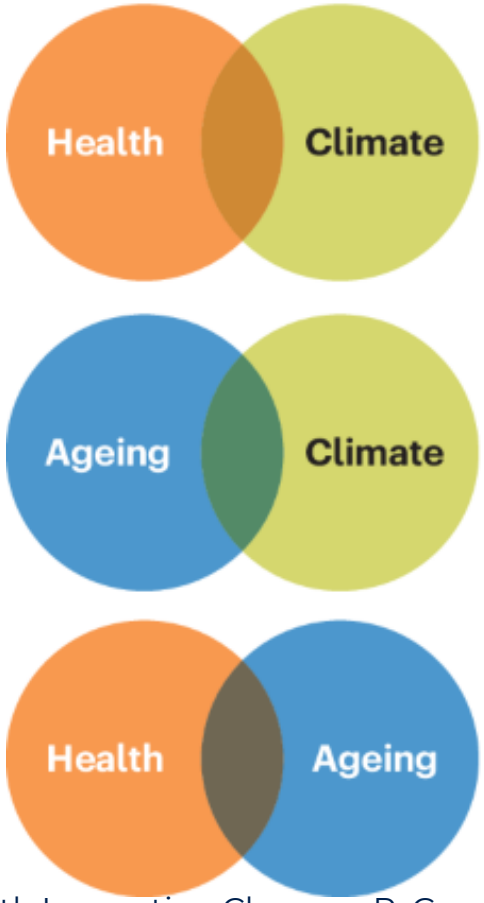
Hallaert, J. (2023). The Fiscal Cost of Aging in Belgium – Pensions and Healthcare. *Selected Issues Papers*, *2023*(065), Article A001, A001. Retrieved Jan 13, 2026

Impacts of climate change on health



https://mcecleanenergy.org/climate-change-and-health-what-you-need-to-know/

Health, ageing, climate



Themes	Study types
• Heat • Thermoregulation • Infectious disease	• Attributions, burdens and health cost • Single events and disasters
• Biophysical vulnerabilities • Air quality • Temperature	• Physiological • Calorimetry • Future projections
• Global trends in Ageing • Chronic conditions • Socio-cultural variation	• Physiological • Socio-demographic

Tipaldo, J.F., Balk, D. & Hunter, L.M. A framework for ageing and health vulnerabilities in a changing climate. *Nat. Clim. Chang.* **14**, 1125–1135 (2024).

The rise of chronic diseases

https://nextlevelmedicalclinic.com/key-chronic-disease-challenges-2025/

WHY INNOVATION MATTERS (especially in Healthcare)

- Health systems face **complex, evolving problems** (aging, climate/health intrication, chronic diseases, workforce pressure, costs)
- Technology alone is **not enough ➔** innovation is about **useful change**, not gadgets.
- Innovation exists at many scales: tools, services, organizations, behaviors.

You are not here to invent "the next big thing,"
but to **make care better, safer, or simpler**.

INNOVATION = A NEW SOLUTION THAT CREATES VALUE FOR REAL USERS

Novel ≠ innovative if it's not useful or adopted.

What innovation is NOT

❌ Just technology

❌ Just ideas

❌ Just startups

In health

- A simple workflow change can be more innovative than AI.
- Adoption and usefulness matter more than sophistication.

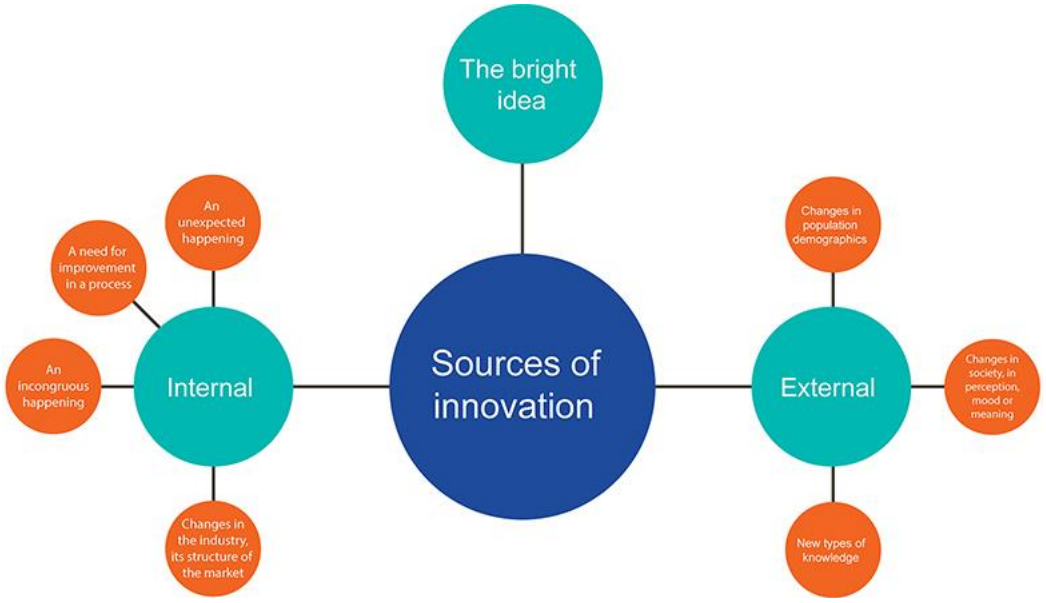
SOURCES OF INNOVATION

Bright ideas but first and foremost:

- Frustrations, inefficiencies, unmet needs
- Observing patients, caregivers, clinicians
- Constraints (time, cost, fatigue, errors)

Many health innovations are born from:

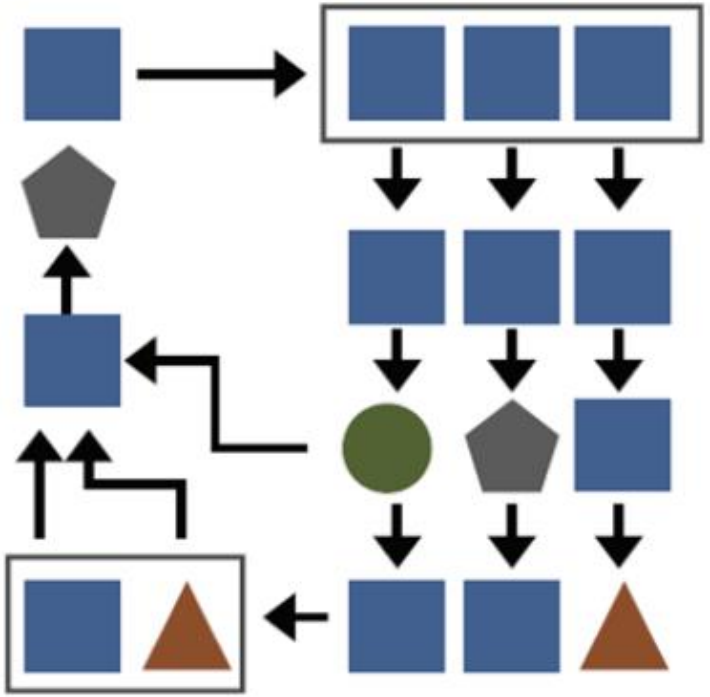
- Workarounds
- “This doesn’t make sense”
- Repeated small failures



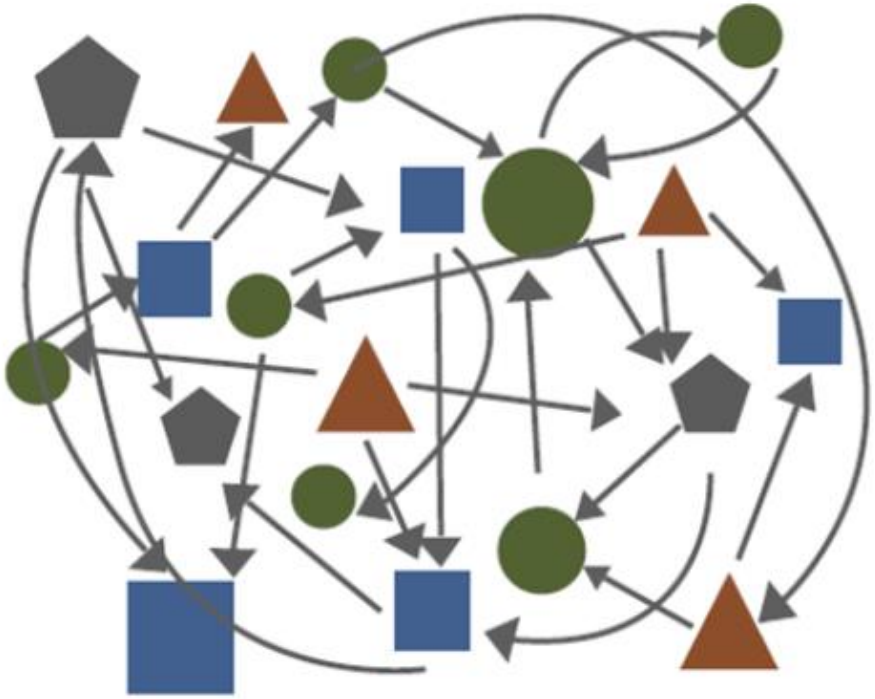
https://www.open.edu/openlearn/mod/oucontent/view.php?id=109136&section=3.1

Design Thinking
Why?

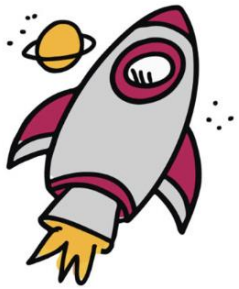
COMPLICATED vs COMPLEX



Complicated



Complex

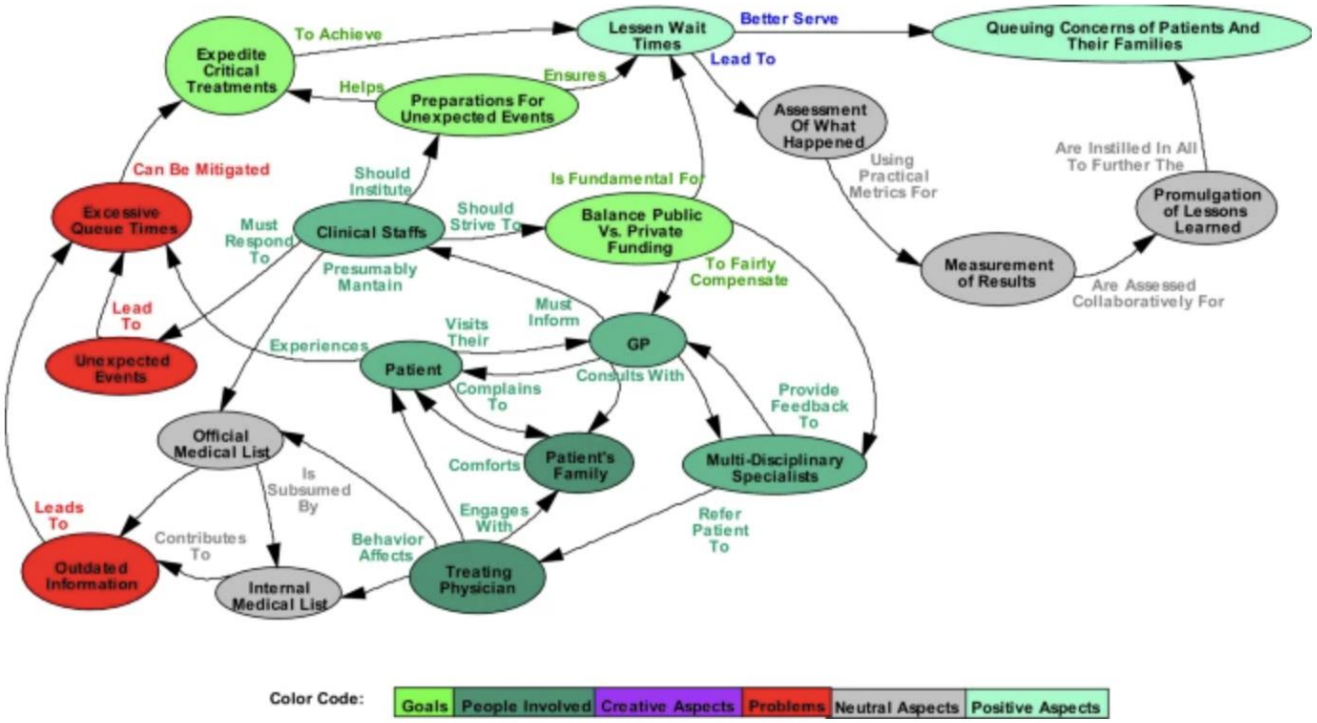
	SIMPLE	COMPLICATED	COMPLEX
	CAKE	ROCKET	CHILD
			
PROBLEM	CLEAR	SOME UNCERTAINTY	LOTS OF UNCERTAINTY
OUTCOMES	Standardized	Most rockets are similiar in important ways	Every child is unique
RULES/ RECIPE	Same rules apply every time	Each effort allows you to improve the recipe for next time	Each child provides experience but adaptation to context is key
EXPERTISE	Not required, but it increases the degree of success	High levels required in specific areas	Can be helpful, but it is neither necessary nor sufficient
PROBABILITY	Follow the recipe for the same product every time	Each test of the recipe improves the probability of success	Rules might help but they might also make things worse

Adapted from Glouberman & Zimmerman

Frameworks Collection by finegood@sfu.ca | Illustrated by sam@drawingchange.com | © CC BY-NC-ND

HEALTHCARE SYSTEM : SIMPLE, COMPLICATED OR COMPLEX?

Fig. 4



The Waiting List Paradox

AN ATTEMPT TO DEFINE DESIGN THINKING

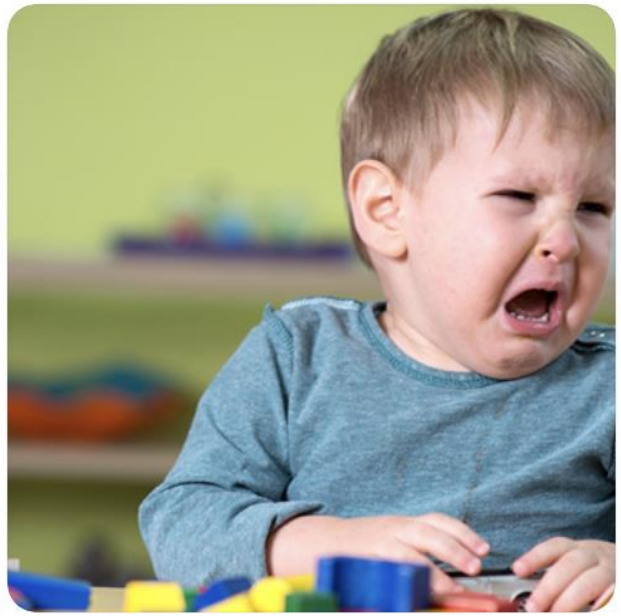
Design Thinking is a combination of a **mindset** and **methods**

It consists in a **human-centered approach** to **innovation** that integrates **desirability**, **feasibility**, and **viability**

ALSO SIMPLE, COMPLICATED AND COMPLEX PROBLEMS

Cake is burnt on the outside, raw inside

Rocket exploded at launch



Child starts to have unpredictable emotional outbursts

CAKE

Why it failed
Oven temperature too high
Baking time incorrect
Wrong pan size

How it is solved
Adjust temperature
Change baking time
Follow the recipe more closely

Key insight
There is a **clear cause**
The fix is **known and repeatable**
Next time, the same fix will likely work again

ROCKET

Why it failed
Sensor malfunction
Software timing error
Structural resonance issue

How it is solved
Post-mortem analysis
Simulations and testing
Redesign specific subsystems

Key insight
Cause–effect exists but is **hard to uncover**
Requires **experts and coordination**
Solutions are not trivial but are **engineerable**

CHILD

Why it failed
Stress at home
Fatigue
Social dynamics
Developmental phase
(Usually **multiple interacting causes**, none sufficient alone.)

How it is addressed
Observe and listen
Try different approaches
Adapt over time
Accept uncertainty

Key insight
No single root cause, **multiple competences required**
Same action may work today and fail tomorrow
“Fixes” are **contextual, provisional, and evolving**

WHY

GENERALLY SPEAKING

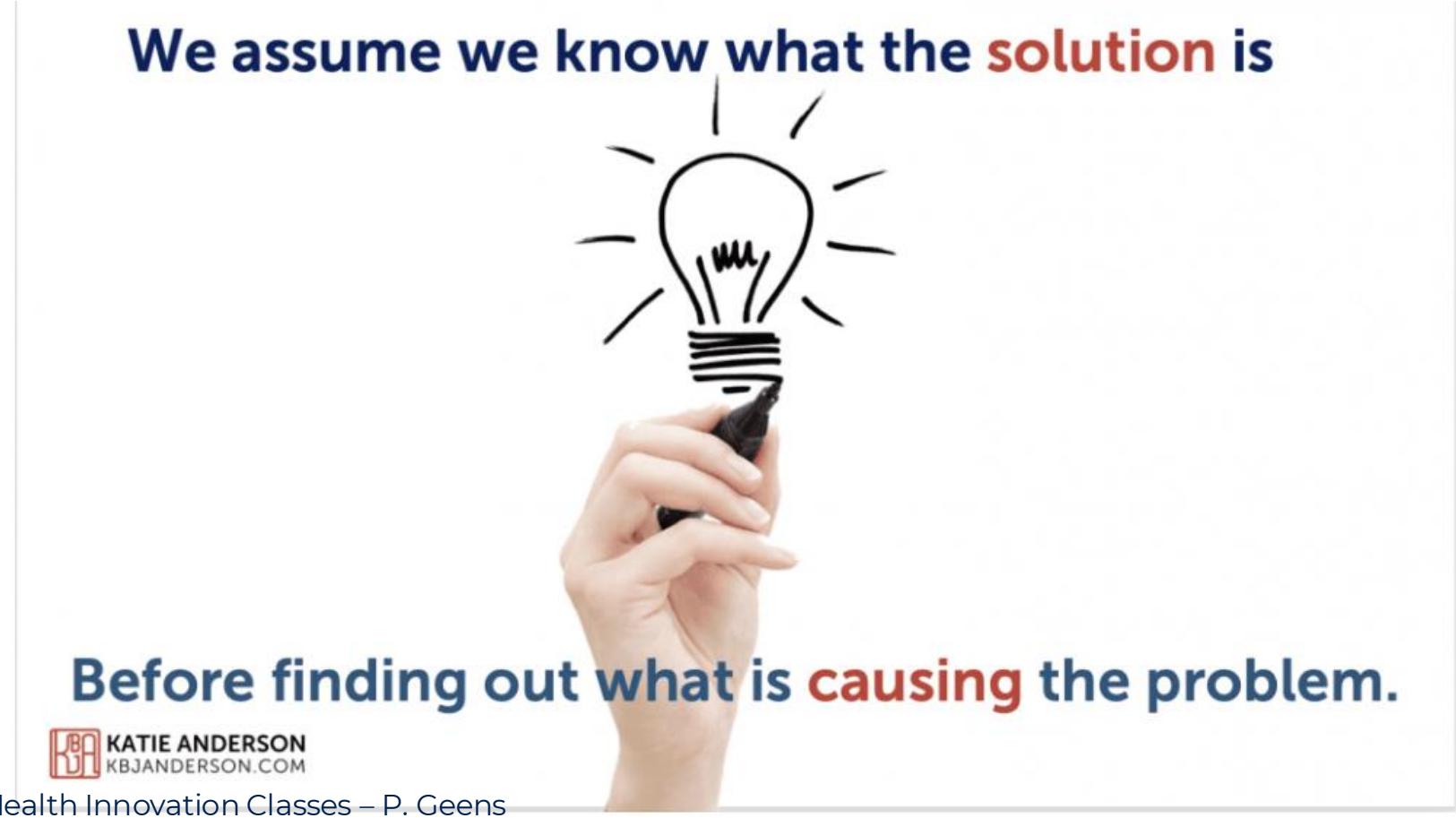
- Many innovation problems are complex, ambiguous, and human-centered
- Traditional linear problem-solving often fails
- Design Thinking helps explore problems before locking solutions

IN HEALTHCARE

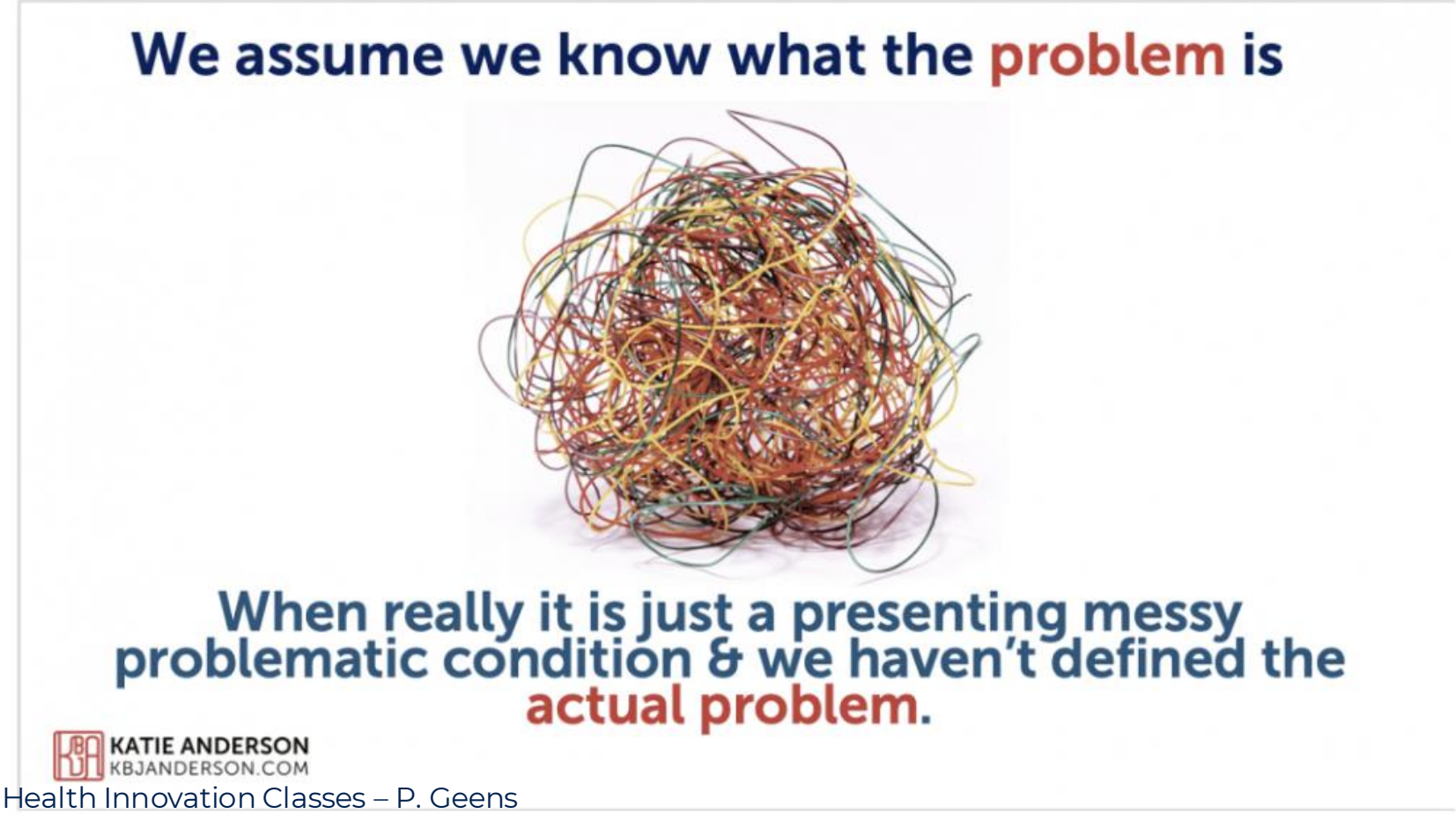
- Healthcare problems involve:
  - Patients,
  - professionals,
  - institutions,
  - Regulations
  - Etc…
- Errors have real human consequences
- Early misunderstanding = high clinical, ethical, and economic cost

Design Thinking
Core principles

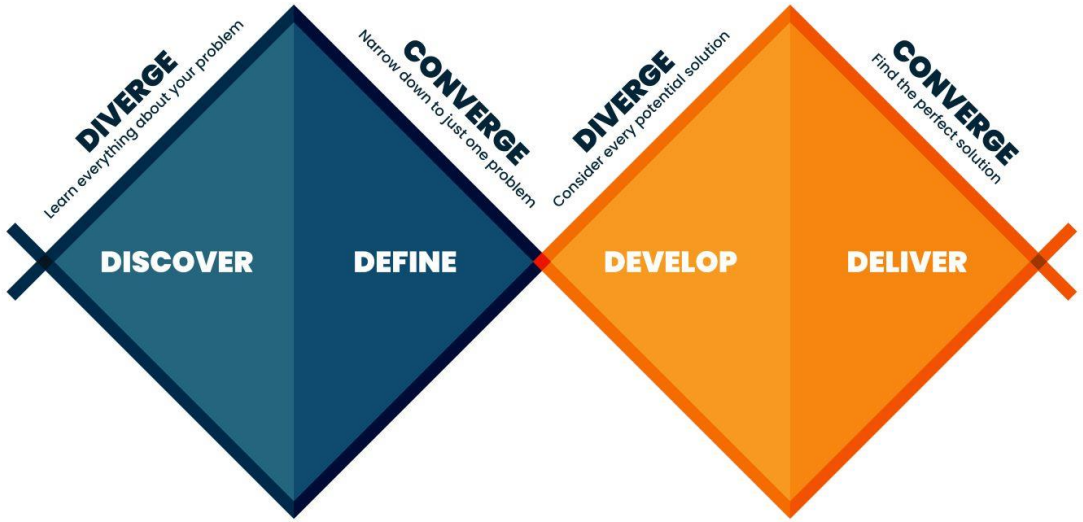
DEFINING THE PROBLEM BEFORE JUMPING ON SOLUTIONS



DEFINING THE **REAL PROBLEM** BEFORE JUMPING ON SOLUTIONS



DIVERGENCE AND CONVERGENCE



https://www.fluxspace.io/resources/the-4-ds-double-diamond-design-thinking-model

- Double Diamond
- Diverge, then converge — twice
- First diamond: understand problem
- Second diamond: explore solutions
- Open wide before narrowing
- Avoid premature problem/solution fixation
- Structure uncertainty and learning

EXPERIMENTATION OVER PERFECTION

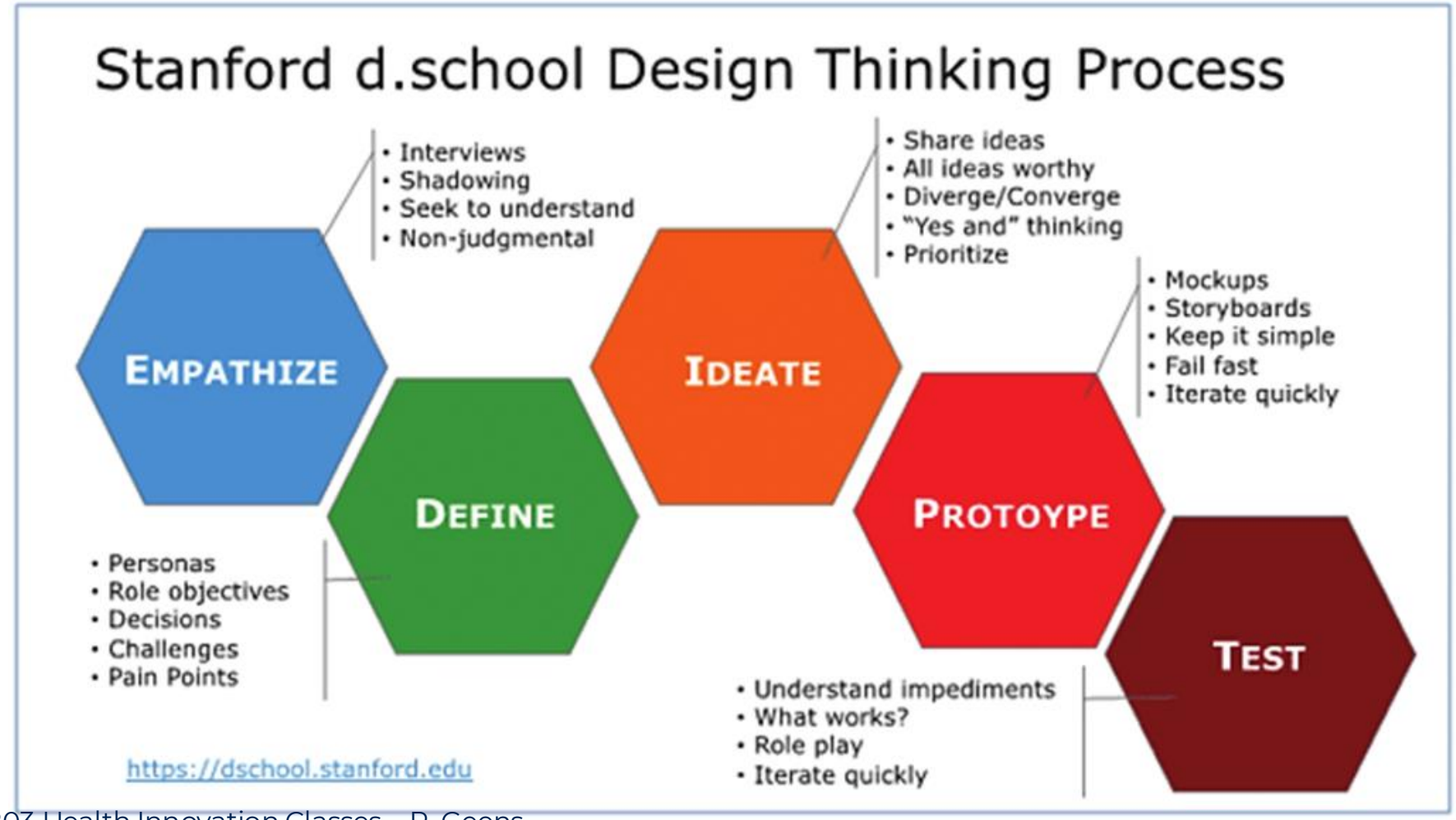
- Learn through action and reflection
- Prototype to explore possibilities
- Feedback guides continuous improvement
- Iteration builds better outcomes
- Small steps enable meaningful change
- Progress emerges from experimentation

ITERATION

- Build, learn, refine, repeat
- Each cycle improves understanding
- Iterate to approach better solutions
- Learning grows through repeated trials
- Refinement emerges from feedback loops
- Progress is cyclical, not linear

Design Thinking Process(es)

DESIGN THINKING PROCESS(ES)



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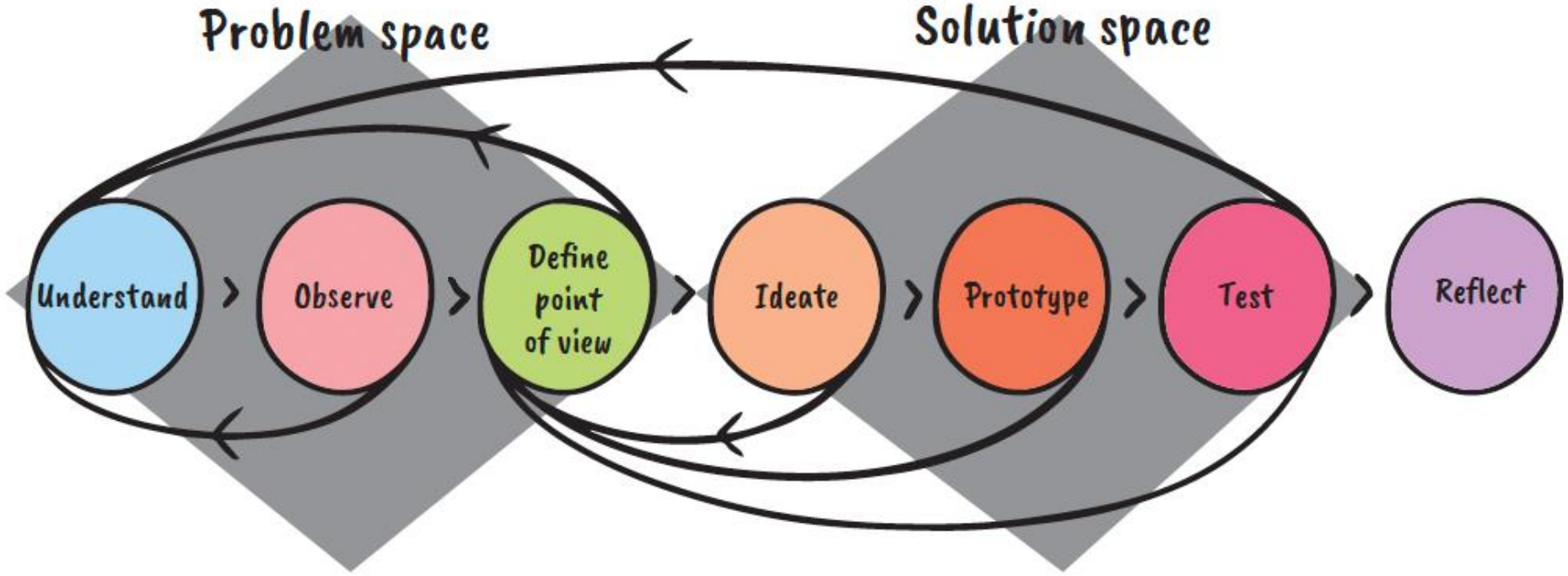
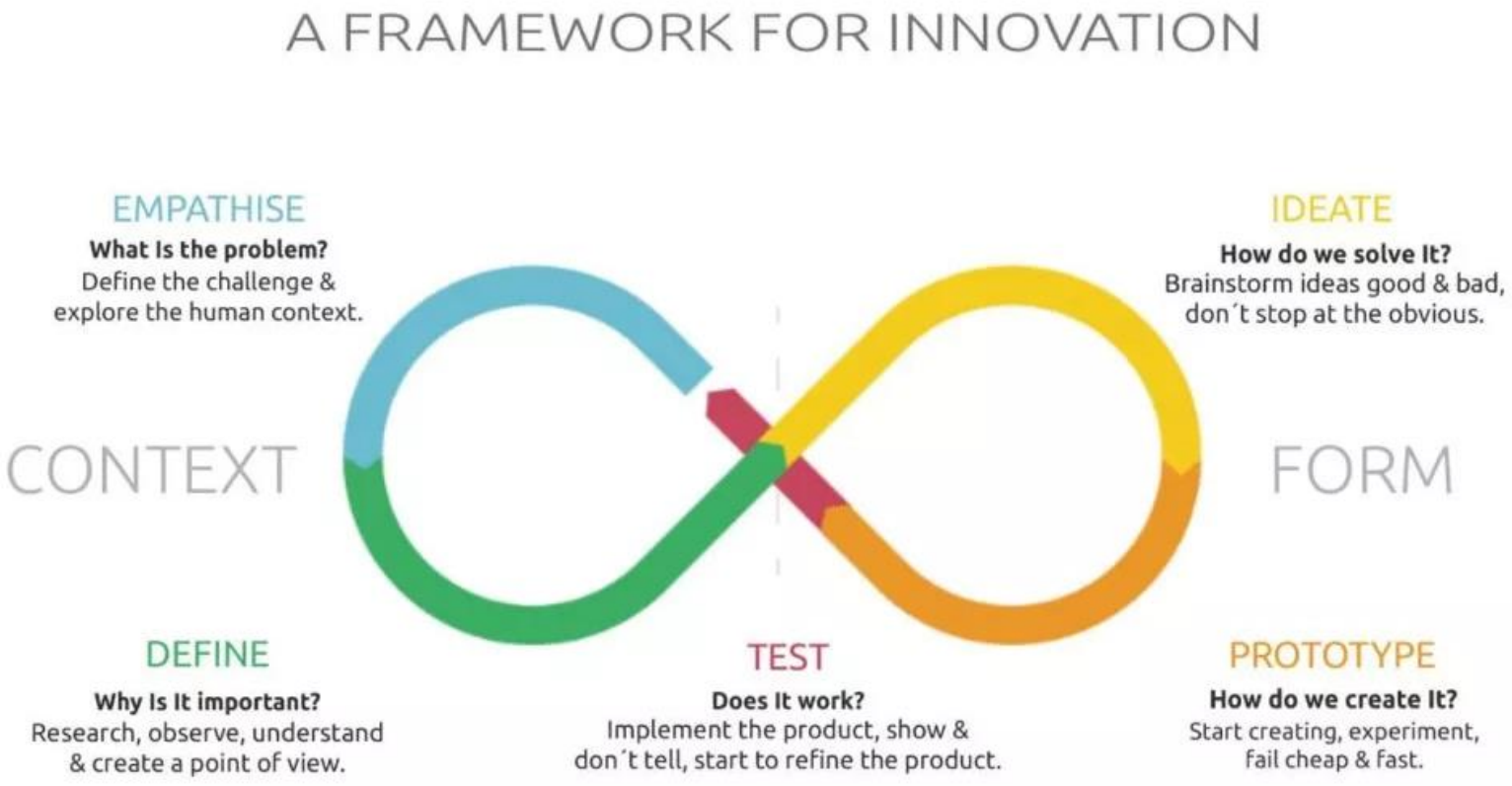


DESIGN THINKING PROCESS(ES)



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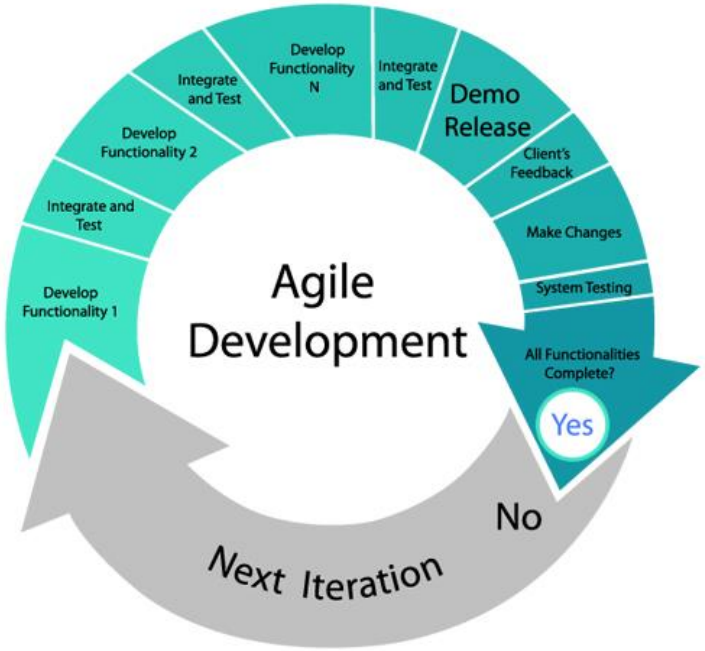
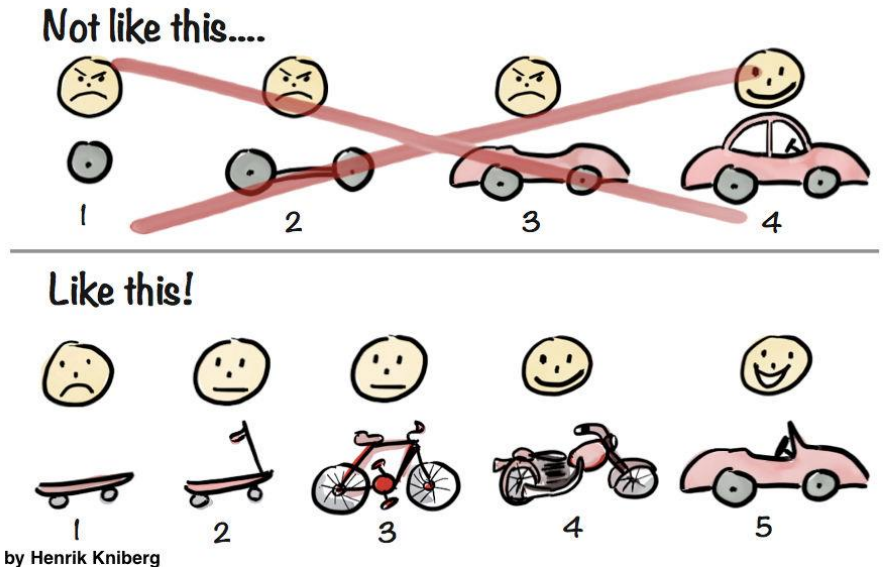
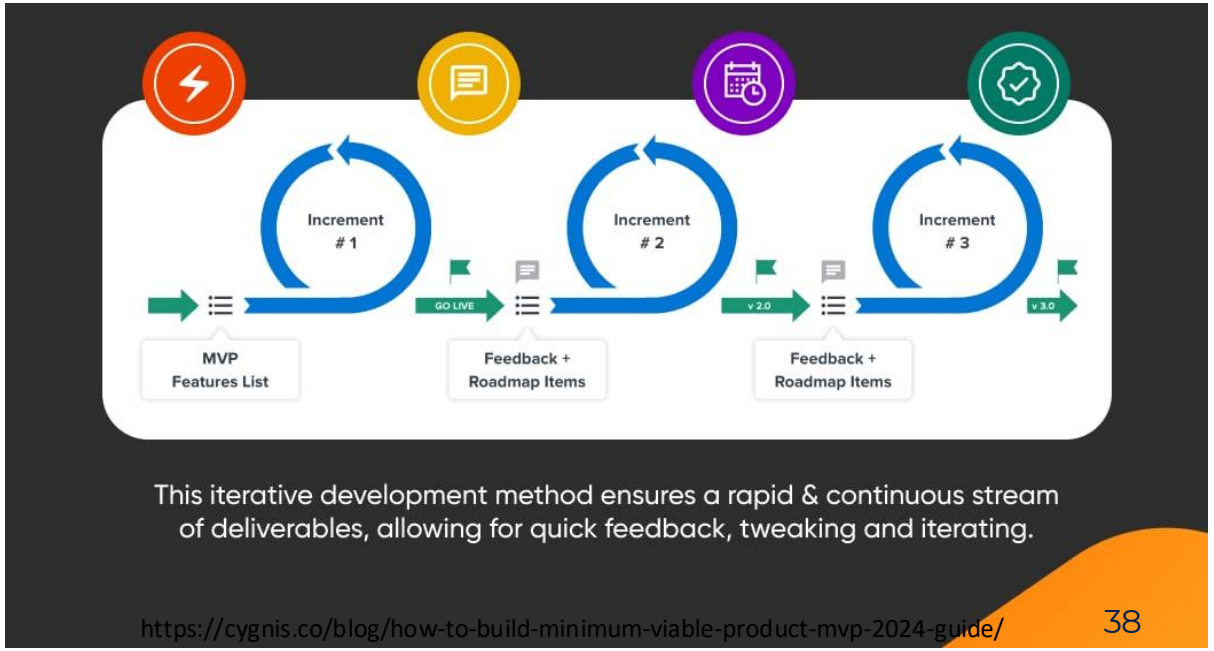
BUT ALSO TO SOME EXTENT



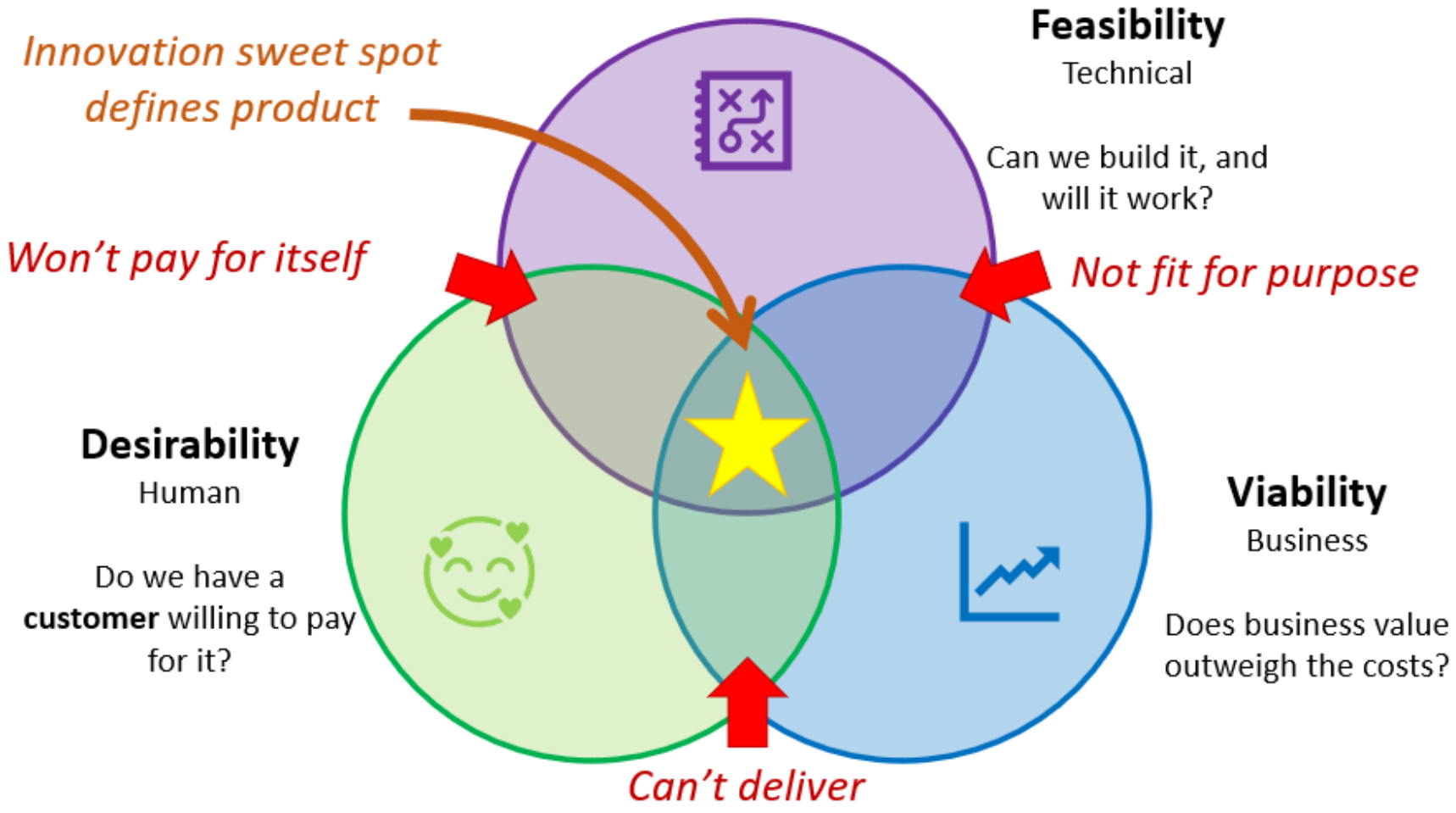




https://nascenia.com/agile-software-development/

https://cygnis.co/blog/how-to-build-minimum-viable-product-mvp-2024-guide/

3 pillars


Innovation sweet spot defines product
Feasibility
Technical
Can we build it, and will it work?
Won't pay for itself
Not fit for purpose
Desirability
Human
Do we have a **customer** willing to pay for it?
Viability
Business
Does business value outweigh the costs?
Can't deliver

Logistics

MOODLE

- Agenda
- Course materials – released week by week
- Assignments
- Communication
- Other useful resources

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AGENDA

2026 février

LUNDI	MARDI	MERCREDI	JEUDI	VENDREDI	SAMEDI	DIMANCHE
26	27	28	29	30	31	01
02	03	04	05	06	07	08
	Videos projets dispos Outil vote dispo		Lancer vote consult #1			
	18h30-20h Intro - Design Thinking - DFV		**18h30-20h cours 2** Empathize Attentes hackathon			
		Visualiser les videos et voter top 3 sur Moodle - Deadline 10/9 23h59				
09	10	11	12	13	14	15
		Assignation groupes et projets				
			12/9 9h - 13/9 12h Hackathon @ GHdC Visites - Hackathon - Annonce groupes			
Visualiser les videos et voter top 3 sur Moodle - Deadline 10/9 23h59			Contact représentants projets - Deadline 18/9 23h59			
16	17	18	19	20	21	22
			Consult #1			
			18h30-21h cours 3 Ecosystème			
Contact représentants projets + planifier ITW - Deadline 18/9 23h59				Stakeholders map + planif RDV + empathizing strategy - Deadline 23/9 23h59		

Posted on Moodle

Contains

- Lectures
- Preparatory activities
- Assignments
- Consulting sessions
- Events

EVALUATION METHOD

- 60% long term = project deliverables
  - 3 intermediary consultations: done - not done
  - Private project defense
  - Final pitch in public

- 20% short term = regular assignments
  - Posted on time on Moodle: done – not done

- 20% individual contribution
  - Self-evaluation and (eventually) peer review using Dynamo

Questions

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